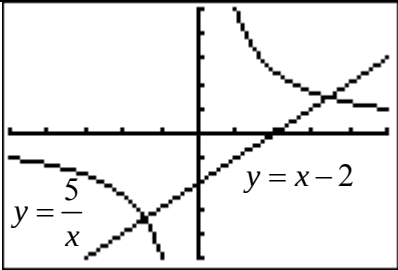


**Anglo-Chinese Junior College**

**H2 Mathematics 9740**

**2009 JC 2 PRELIM Solution**

**Paper 1:**

|                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |  |
|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| <b>1</b>       |  <p>Solving for point of intersection:</p> $x - 2 = \frac{5}{x} \Rightarrow x^2 - 2x - 5 = 0 \Rightarrow x = \frac{2 \pm \sqrt{4 + 20}}{2} = 1 \pm \sqrt{6}$ <p>For <math>x - 2 \geq \frac{5}{x}</math>, <math>1 - \sqrt{6} \leq x &lt; 0</math> or <math>x \geq 1 + \sqrt{6}</math></p>                                                                                                                           |  |
|                | <p>Using previous results,</p> <p>For <math> x  - 2 &gt; \frac{5}{ x }</math>, <math>1 - \sqrt{6} &lt;  x  &lt; 0</math> (rej.) or <math> x  &gt; 1 + \sqrt{6}</math></p> <p><math>\therefore x &lt; -(1 + \sqrt{6})</math> or <math>x &gt; 1 + \sqrt{6}</math></p>                                                                                                                                                                                                                                 |  |
| <b>2</b>       | $\sum_{r=1}^n (U_{r+1} - U_r) = \sum_{r=1}^n ((e-1)e^{r+1})$ $U_{n+1} - U_1 = (e-1) \sum_{r=1}^n (e^{r+1})$ $U_{n+1} = (e-1) \frac{e^2(e^n - 1)}{e - 1} + e^2 = e^{n+2}$ $U_n = e^{n+1}$                                                                                                                                                                                                                                                                                                            |  |
|                | $U_2 U_3 U_4 \dots U_n = e^3 \cdot e^4 \cdot e^5 \dots e^{n+1}$ $= e^{\frac{1}{2}(n-1)(n+4)}$                                                                                                                                                                                                                                                                                                                                                                                                       |  |
| <b>3</b>       | <p>Let <math>x</math>, <math>y</math> and <math>z</math> be the cost of the flights for each parent, each grandparent and each child respectively.</p> $2x + 2y + z = 1959 \text{ ----- (1)}$ $x + 2y + 3z = 2196 \text{ ----- (2)}$ $2x + y + 4z = 2640 \text{ ----- (3)}$ $x + y + 2z = ?$ <p>Solving (1), (2) &amp; (3) using GC,<br/> <math>x = 453</math>, <math>y = 354</math> and <math>z = 345</math><br/>         Cost for Tan family = <math>\\$[453 + 354 + 2(345)] = \\$1497</math></p> |  |
| <b>4 (i)</b>   | $S_1 = \frac{1}{2}, \quad S_2 = \frac{2}{3}, \quad S_3 = \frac{3}{4}, \quad S_4 = \frac{4}{5}$                                                                                                                                                                                                                                                                                                                                                                                                      |  |
| <b>4 (ii)</b>  | $f(1) = \frac{1}{2}, \quad f(2) = \frac{1}{3}, \quad f(3) = \frac{1}{4}, \quad f(4) = \frac{1}{5}$ $\therefore f(n) = \frac{1}{n+1}$                                                                                                                                                                                                                                                                                                                                                                |  |
| <b>4 (iii)</b> | <p>Let <math>P_n</math> denote the statement <math>S_n = 1 - \frac{1}{n+1}</math>, <math>\forall n \in \mathbb{N}^+</math>.</p>                                                                                                                                                                                                                                                                                                                                                                     |  |

For  $n = 1$ ,  $\text{LHS} = S_1 = \frac{1}{2}$   $\text{RHS} = 1 - \frac{1}{1+1} = \frac{1}{2}$   
 $\therefore \text{LHS} = \text{RHS}.$   
 $\therefore P_1$  is true.

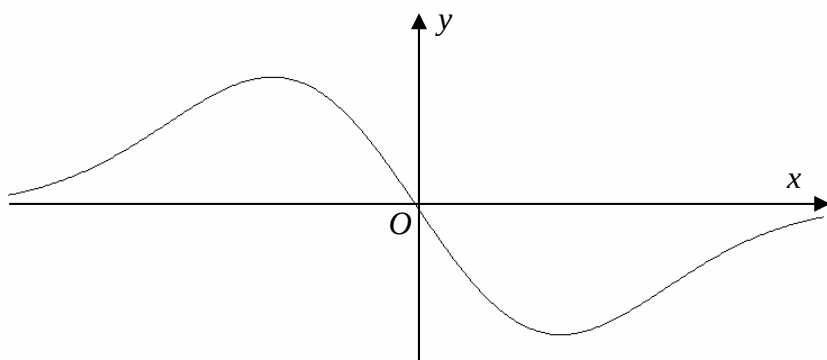
Assume  $P_k$  true for some  $k \in \mathbb{N}^+$ , i.e.  $S_k = 1 - \frac{1}{k+1}$

Prove that  $P_{k+1}$  is true, i.e.  $S_{k+1} = 1 - \frac{1}{k+2}$

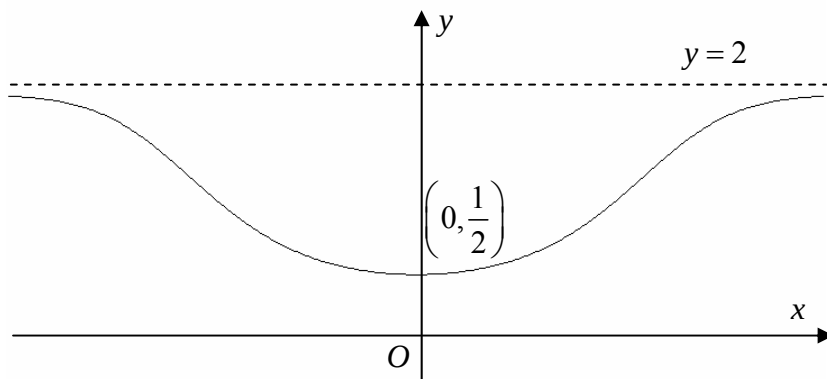
$$\begin{aligned} \text{LHS} = S_{k+1} &= S_k + \frac{1}{(k+1)(k+2)} \\ &= 1 - \frac{1}{k+1} + \frac{1}{(k+1)(k+2)} \\ &= 1 - \frac{k+1}{(k+1)(k+2)} \\ &= 1 - \frac{1}{k+2} = \text{RHS} \end{aligned}$$

$\therefore P_{k+1}$  is true if  $P_k$  is true. But  $P_1$  is true.  
 $\therefore$  By the principle of mathematical induction,  $P_n$  is true  $\forall n \in \mathbb{N}^+.$

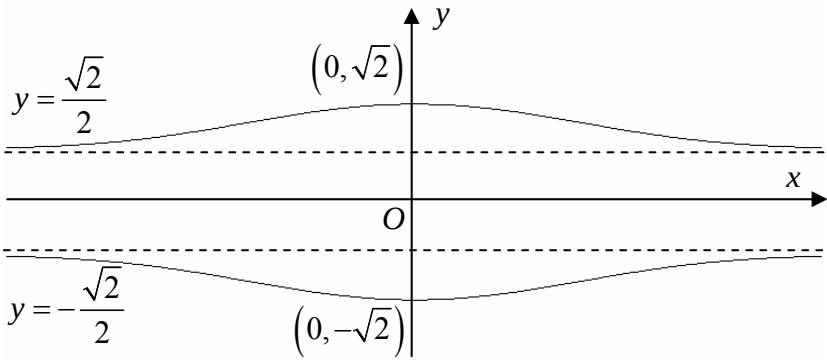
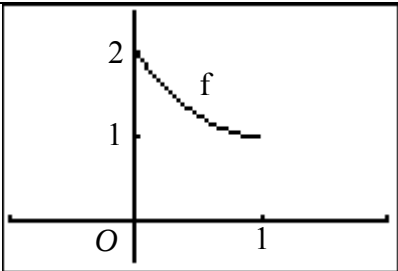
5 (i)  $y = f'(x)$



5 (ii)  $y = \frac{1}{f(x)}$



5 (iii)  $y^2 = f(x)$

|         |                                                                                                                                                                                                                                                                                                                                                  |  |
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|         |                                                                                                                                                                                                                                                                |  |
| 6 (i)   |  <p>Since <math>f</math> is one-one for <math>0 \leq x \leq 1</math>, <math>f^{-1}</math> exists.</p> $y = x^2 - 2x + 2$ $= (x-1)^2 + 1 \Rightarrow x = 1 - \sqrt{y-1} \because 0 \leq x \leq 1$ $\therefore f^{-1}(x) = 1 - \sqrt{x-1}, \quad 1 \leq x \leq 2$ |  |
| 6 (ii)  | <p>Range of <math>f = [1, 2]</math>      Domain of <math>g = [0, 3]</math></p> <p>Since <math>R_f \subseteq D_g</math>, <math>gf</math> exists.</p> $gf(x) = \frac{x^2 - 2x + 4}{x^2 - 2x + 3}, \quad 0 \leq x \leq 1$                                                                                                                           |  |
| 6 (iii) | <p>Restricted Domain of <math>g = [1, 2]</math></p> <p>Range of <math>gf = \left[\frac{4}{3}, \frac{3}{2}\right]</math></p>                                                                                                                                                                                                                      |  |
| 7 (i)   | <p>Differentiating <math>\frac{1}{x} + \frac{1}{y} = \frac{1}{a}</math> w.r.t. <math>x</math>,</p> $-\frac{1}{x^2} - \frac{1}{y^2} \frac{dy}{dx} = 0$ $\frac{dy}{dx} = -\frac{y^2}{x^2}$ <p>Since <math>\frac{dy}{dx} &lt; 0</math>, <math>y</math> is a decreasing function.</p>                                                                |  |
| 7 (ii)  | <p>There are no stationary points on the curve because <math>\frac{dy}{dx} \neq 0</math> as <math>y \neq 0</math></p>                                                                                                                                                                                                                            |  |
| 7 (iii) | <p>Equation of tangent is <math>\frac{y-2a}{x-2a} = -\frac{(2a)^2}{(2a)^2} - 1</math></p>                                                                                                                                                                                                                                                        |  |

Therefore,  $y = -x + 4a$

Substituting  $y = -x + 4a$  into eqn of curve

$$\frac{1}{x} + \frac{1}{4a - x} = \frac{1}{a}$$

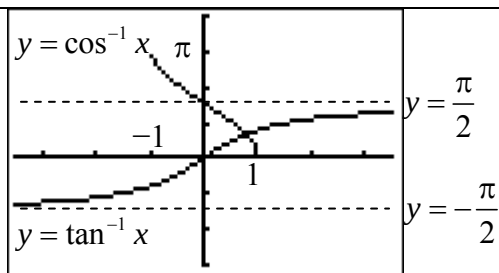
$$\Rightarrow x^2 - 4ax + 4a^2 = 0$$

$$\Rightarrow (x - 2a)^2 = 0$$

$$\Rightarrow x = 2a$$

Since only one value of  $x$  is obtained,  
the tangent does not meet the curve again.

**8 (a)**



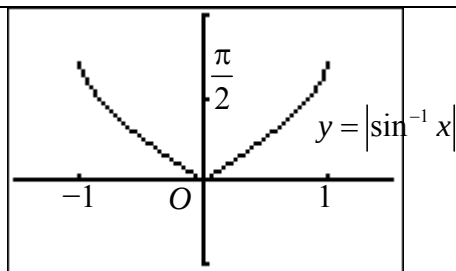
Domain of  $\cos^{-1} x$ :  $-1 \leq x \leq 1$

Range of  $\cos^{-1} x$ :  $0 \leq \cos^{-1} x \leq \pi$

Domain of  $\tan^{-1} x$ :  $-\infty < x < \infty$

Range of  $\tan^{-1} x$ :  $-\frac{\pi}{2} < \tan^{-1} x < \frac{\pi}{2}$

**8 (b)**



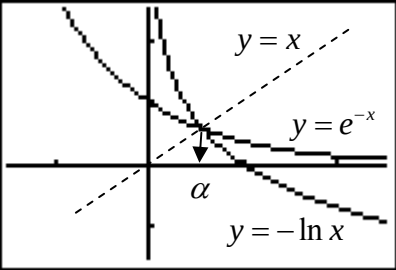
$$|\sin^{-1} x| = \frac{\pi}{3} \Rightarrow \sin^{-1} x = \pm \frac{\pi}{3}$$

$$\Rightarrow x = \sin\left(-\frac{\pi}{3}\right) \text{ or } x = \sin\left(\frac{\pi}{3}\right)$$

$$\therefore x = \pm \frac{\sqrt{3}}{2}$$

**8 (c)**

The graph of  $y = e^{-x}$  can be obtained from the graph of  $y = -\ln x$  by a reflection in the line  $y = x$ .

|                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |  |
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|                |  <p>For <math>-\ln x &gt; e^{-x}</math>, <math>0 &lt; x &lt; \alpha</math></p>                                                                                                                                                                                                                                                                                                                                              |  |
| <b>9 (i)</b>   | $u = x + y \Rightarrow \frac{du}{dx} = 1 + \frac{dy}{dx}$ <p>d.e. becomes <math>\frac{du}{dx} - 2 - u^2 = 0</math> after replacing <math>\frac{dy}{dx}</math> and <math>x + y</math></p> <p>Therefore, <math>\frac{du}{dx} = 2 + u^2</math></p>                                                                                                                                                                                                                                                             |  |
| <b>9 (ii)</b>  | $\int \frac{1}{2 + u^2} du = \int dx$ $\Rightarrow \frac{1}{\sqrt{2}} \tan^{-1} \left( \frac{u}{\sqrt{2}} \right) = x + C$ $\Rightarrow \frac{u}{\sqrt{2}} = \tan(\sqrt{2}x + D) \text{ where } D = \sqrt{2}C$ $\Rightarrow y = \sqrt{2} \tan(\sqrt{2}x + D) - x$ <p>When <math>x = 0</math>, <math>y = 2</math>. Therefore, <math>2 = \sqrt{2} \tan D</math>. <math>D = \tan^{-1} \sqrt{2}</math></p> <p>The particular solution is <math>y = \sqrt{2} \tan(\sqrt{2}x + \tan^{-1} \sqrt{2}) - x</math></p> |  |
| <b>9 (iii)</b> | $A = x(\sqrt{2} \tan(\sqrt{2}x + \tan^{-1} \sqrt{2}) - x)$ $\frac{dA}{dz} = \frac{d(xy)}{dz} = x \frac{dy}{dz} + y \frac{dx}{dz}$ <p>When <math>x = 0</math>, <math>y = 2</math> therefore, <math>\frac{dx}{dz} = \frac{3}{2}</math></p>                                                                                                                                                                                                                                                                    |  |
| <b>10 (a)</b>  | $z + (2 + i)w = -9 + 16i \quad (1)$ $z^* + w = 3i \quad (2)$ <p>Substitute <math>w = 3i - z^*</math> into equation (1)</p> $z + (2 + i)(3i - z^*) = -9 + 16i$ $z + (-2 - i)z^* + (-3 + 6i) = -9 + 16i$ $z + (-2 - i)z^* = -6 + 10i$ <p>Let <math>z = x + iy</math></p> $(x + iy) + (-2 - i)(x - iy) = -6 + 10i$ $(-x - y) + i(-x + 3y) = -6 + 10i$                                                                                                                                                          |  |

|                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |  |
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|                | <p>Equating real parts: <math>-x - y = -6 \Leftrightarrow x + y = 6</math> (3)</p> <p>Equating imaginary parts: <math>-x + 3y = 10</math> (4)</p> <p>Solving equations (3) and (4): <math>x = 2</math> and <math>y = 4</math></p> <p><math>z = 2 + 4i</math></p> <p><math>w = 3i - (2 - 4i) = -2 + 7i</math></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |  |
| <b>10 (b)</b>  | <p><math>z^3 = i</math></p> <p><math>z^3 = \exp\{i(2n\pi + \frac{\pi}{2})\}</math> where <math>n \in \mathbb{Z}</math></p> <p><math>z = \exp\{i(\frac{2n\pi}{3} + \frac{\pi}{6})\}</math></p> <p><math>n = -1, z = \exp\{i(-\frac{\pi}{2})\}</math></p> <p><math>n = 0, z = \exp\{i(\frac{\pi}{6})\}</math></p> <p><math>n = 1, z = \exp\{i(\frac{5\pi}{6})\}</math></p> <p><math>w^3 = -i</math></p> <p><math>(w^3)^* = (-i)^*</math></p> <p><math>(w^*)^3 = i</math></p> <p>From the results of (a) above,</p> <p><math>w^* = \exp\{i(-\frac{\pi}{2})\}</math></p> <p><math>w^* = \exp\{i(\frac{\pi}{6})\}</math></p> <p><math>w^* = \exp\{i(\frac{5\pi}{6})\}</math></p> <p>Hence <math>w = \exp\{i\frac{\pi}{2}\}, \exp\{i(-\frac{\pi}{6})\}, \exp\{i(-\frac{5\pi}{6})\}</math></p> |  |
| <b>11 (i)</b>  | <p><math>y = \cot\left(2x + \frac{\pi}{4}\right)</math></p> <p><math>\Rightarrow \frac{dy}{dx} = -\operatorname{cosec}^2\left(2x + \frac{\pi}{4}\right) \cdot 2 = -2(1 + y^2)</math></p> <p><math>\frac{d^2y}{dx^2} = -4y \frac{dy}{dx}</math></p> <p><math>\Rightarrow \frac{d^3y}{dx^3} = -4\left[y \frac{d^2y}{dx^2} + \left(\frac{dy}{dx}\right)^2\right]</math>. Therefore, <math>k = -4</math></p>                                                                                                                                                                                                                                                                                                                                                                                |  |
| <b>11 (ii)</b> | <p>When <math>x = 0, y = 1, \frac{dy}{dx} = -4, \frac{d^2y}{dx^2} = 16, \frac{d^3y}{dx^3} = -128</math></p> <p>Therefore, <math>\cot\left(2x + \frac{\pi}{4}\right) \approx 1 - 4x + 8x^2 - \frac{64}{3}x^3</math></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |  |

|          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |  |
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| 11 (iii) | <p>Differentiating the expansion above,</p> $-2 \operatorname{cosec}^2 \left( 2x + \frac{\pi}{4} \right) \approx -4 + 16x - 64x^2$ <p>Therefore, <math>\operatorname{cosec}^2 \left( 2x + \frac{\pi}{4} \right) \approx 2 - 8x + 32x^2</math></p> <p>Let <math>2x + \frac{\pi}{4} = \frac{13\pi}{50}</math>. Then <math>x = \frac{\pi}{200}</math></p> <p>Hence, <math>\operatorname{cosec}^2 \left( \frac{13\pi}{50} \right) = 2 - \frac{\pi}{25} + \frac{\pi^2}{1250}</math> where</p> $a = 2, b = -\frac{1}{25}, c = \frac{1}{1250}$                                                                                                                                                                                                                                                                                                                                                                                                                                                        |  |
| 12 (i)   | <p>Let N be the foot of perpendicular from A to the plane <math>p_1</math>.</p> $\overrightarrow{AN} = \lambda \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$ $\overrightarrow{ON} = \overrightarrow{OA} + \overrightarrow{AN} = \begin{pmatrix} -1 \\ 2 \\ 4 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} -1 + \lambda \\ 2 + \lambda \\ 4 + \lambda \end{pmatrix}$ <p>Since N is a point on the plane <math>p_1</math>,</p> $\overrightarrow{ON} \cdot \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = 7$ $\begin{pmatrix} -1 + \lambda \\ 2 + \lambda \\ 4 + \lambda \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = 7$ $(-1 + \lambda) + (2 + \lambda) + (4 + \lambda) = 7$ $\lambda = 2$ $\overrightarrow{ON} = \begin{pmatrix} -1 + \lambda \\ 2 + \lambda \\ 4 + \lambda \end{pmatrix} = \begin{pmatrix} -1 + \frac{2}{3} \\ 2 + \frac{2}{3} \\ 4 + \frac{2}{3} \end{pmatrix} = \frac{1}{3} \begin{pmatrix} -1 \\ 8 \\ 14 \end{pmatrix}$ |  |
| 12 (ii)  | $\begin{pmatrix} 4 \\ 3 \\ 0 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = 4 + 3 + 0 = 7 \text{ (satisfies equation of plane } p_1 \text{)}$ $\begin{pmatrix} 4 \\ 3 \\ 0 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ -1 \\ a \end{pmatrix} = 4 - 3 + 0 = 1 \text{ (satisfies equation of plane } p_2 \text{)}$ <p>Hence the point (4, 3, 0) is on both planes <math>p_1</math> and <math>p_2</math>.</p> <p>(4, 3, 0) is then a point on the line <math>\ell</math> where the 2 planes</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |  |

|          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |  |
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|          | <p>intersect.</p> $\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \times \begin{pmatrix} 1 \\ -1 \\ a \end{pmatrix} = \begin{pmatrix} 1+a \\ 1-a \\ -2 \end{pmatrix}$ <p>is the direction vector of the line <math>\ell</math>,</p> <p>Hence <math>\mathbf{r} = \begin{pmatrix} 4 \\ 3 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} 1+a \\ 1-a \\ -2 \end{pmatrix}</math> is the equation of line <math>\ell</math>.</p>                                                                                                                                                                                                                                                              |  |
| 12 (iii) | <p>If the planes <math>p_3</math> with equation <math>\mathbf{r} \cdot \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} = b</math> intersects with <math>p_1</math> and <math>p_2</math> at line <math>\ell</math>, then</p> $\begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} \cdot \begin{pmatrix} 1+a \\ 1-a \\ -2 \end{pmatrix} = 0$ $(1+a) + 2(1-a) - 6 = 0$ $a = -3$ <p>The line <math>\ell</math> lies in the plane <math>p_3</math> and thus <math>\begin{pmatrix} 4 \\ 3 \\ 0 \end{pmatrix}</math> lies in the plane <math>p_3</math></p> <p>also.</p> $\begin{pmatrix} 4 \\ 3 \\ 0 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} = b \Rightarrow b = 4 + 6 + 0 = 10$ |  |
| 12 (iv)  | $\overline{AB} = \overline{OB} - \overline{OA} = \begin{pmatrix} 0 \\ -2 \\ 5 \end{pmatrix} - \begin{pmatrix} -1 \\ 2 \\ 4 \end{pmatrix} = \begin{pmatrix} 1 \\ -4 \\ 1 \end{pmatrix}.$ <p>Given the length of projection of line segment AB on the line <math>\ell</math> is <math>\frac{1}{\sqrt{6}}</math>.</p> $\frac{\left  \overline{AB} \cdot \begin{pmatrix} 1+a \\ 1-a \\ -2 \end{pmatrix} \right }{\left  \begin{pmatrix} 1+a \\ 1-a \\ -2 \end{pmatrix} \right } = \frac{1}{\sqrt{6}}$ $\frac{ (1+a) - 4(1-a) - 2 }{\sqrt{(1+a)^2 + (1-a)^2 + (-2)^2}} = \frac{1}{\sqrt{6}}$ $\frac{ 5(a-1) }{\sqrt{(a^2+3)}} = \frac{1}{\sqrt{3}}$                                   |  |



|                                                                                                                                                                                                 |  |
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| $\frac{25(a-1)^2}{(a^2+3)} = \frac{1}{3}$ $75(a-1)^2 = a^2 + 3$ $74a^2 - 150a + 72 = 0$ $a = \frac{150 \pm \sqrt{1188}}{2(74)} = \frac{75 \pm \sqrt{247}}{74}$ $a = 0.781 \text{ or } a = 1.25$ |  |
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